

TE Curve Verification Pipeline Multi-Model Curve Comparison Report

Feedforward Transmission Error Baseline

Styled PDF edition generated from the canonical Markdown analysis report for improved readability, section hierarchy, and table presentation.

Overview

This report compares representative TE Curve Verification Pipeline TE curves by overlaying multiple model predictions on the same original measured curve. The plots are intended to show whether each model tracks the local harmonic oscillations rather than only the broad mean trend.

Scope

- each comparison image contains four deterministic held-out test curves;
- forward comparisons are shown on forward curves only;
- backward comparisons are shown on backward curves only;
- Wave 1 screening keeps the three strongest family-best models by Curve MAE [deg] within each direction;
- Original Curve uses the same visual weight as predictions and a dark-gray color for balanced comparison.

Metrics Summary

Forward Reference Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_original_best_Fw	rcim_original	Fw	0.002769	0.002951	6.250
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819

Forward Wave 1 Family Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
feedforward_fw	wave1_current_registry	Fw	0.086931	0.087006	190.832
harmonic_regression_fw	wave1_current_registry	Fw	0.003230	0.003494	7.185
periodic_mlp_fw	wave1_current_registry	Fw	0.089150	0.089207	195.822
residual_harmonic_mlp_fw	wave1_current_registry	Fw	0.085646	0.085696	187.950
tree_fw	wave1_current_registry	Fw	0.066315	0.066379	145.420
periodic_mlp_harmonic_fw	wave1_periodic_mlp_harmonic_campaign	Fw	0.003254	0.003553	7.232

Forward Wave 2.1 Temporal Model Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.088705	0.088763	195.075
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.087172	0.087229	191.682
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.088000	0.088052	193.497
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.085364	0.085443	187.236
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.086270	0.086329	189.713
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.087733	0.087808	192.894

Forward Wave 2.3 Residual Harmonic Temporal Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
residual_harmonic_gru_sequence_sparse_rcim_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003194	0.003499	7.083
residual_harmonic_gru_sequence_dense240_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.006983	0.009275	15.722
residual_harmonic_gru_sequence_dense360_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007869	0.010574	17.740
residual_harmonic_lstm_sequence_sparse_rcim_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003229	0.003533	7.164
residual_harmonic_lstm_sequence_dense240_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007042	0.009370	15.868
residual_harmonic_lstm_sequence_dense360_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007731	0.010235	17.430

Forward Wave 3.1 Offset-Aware Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_1_sequential_residual_offset_probe_fw	wave3_1_offset_aware_probe_registry	Fw	0.003377	0.003799	7.487

Forward Wave 3.2 Harmonic-Offset Probe Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_2_clean_sequential_residual_offset_fw	wave3_2_harmonic_offset_probe_registry	Fw	0.003439	0.003870	7.632
wave3_2_harmonic_residual_offset_fw	wave3_2_harmonic_offset_probe_registry	Fw	0.002850	0.003108	6.286

Forward Wave 3.3 Curve-Aware Training Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave3_3_curve_aware_pointwise_control_fw	wave3_3_curve_aware_training_registry	Fw	0.003362	0.003612	7.474
wave3_3_raw_centered_shape_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.003174	0.003429	7.047
wave3_3_raw_offset_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.003269	0.003588	7.268
wave3_3_full_curve_composite_fw	wave3_3_curve_aware_training_registry	Fw	0.003251	0.003515	7.209

Forward Wave 4.1 Robust-Loss Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_1_mae_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003134	0.003382	6.956
wave4_1_smooth_l1_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003300	0.003545	7.342
wave4_1_log_cosh_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003344	0.003595	7.427

Forward Wave 4.2 Quantile Probabilistic Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_2_quantile_p10_p50_p90_fw	wave4_2_probabilistic_registry	Fw	0.003276	0.003545	7.279
wave4_2_gaussian_nll_fw	wave4_2_probabilistic_registry	Fw	0.003156	0.003415	7.008

Forward Wave 4.3 Mixture Density Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_3_mixture_density_k2_fw	wave4_3_mixture_density_registry	Fw	0.003329	0.003593	7.388
wave4_3_mixture_density_k3_fw	wave4_3_mixture_density_registry	Fw	0.003226	0.003487	7.164

Forward Wave 4.4 Latent State Hysteresis Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave4_4_gru_latent_offset_residual_fw	wave4_4_latent_state_hysteresis_registry	Fw	0.003549	0.003996	7.873
wave4_4_causal_tcn_latent_offset_residual_fw	wave4_4_latent_state_hysteresis_registry	Fw	0.003476	0.003939	7.717

Forward Wave 3 Harmonic Prior Residual Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave5_1_harmonic_prior_pointwise_control_fw	wave5_1_harmonic_prior_residual_registry	Fw	0.003374	0.003655	7.501
wave5_1_harmonic_prior_smooth_l1_structured_fw	wave5_1_harmonic_prior_residual_registry	Fw	0.003514	0.003768	7.812

Forward Wave52b Offset Harmonic Guided Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
wave52b_offset_centered_shape_harmonic_fw	wave52b_offset_harmonic_guided_registry	Fw	0.058653	0.058748	128.839

Forward Polished Model Development Registry Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
polished_feedforward_Fw	polished_model_development_registry	Fw	0.086931	0.087006	190.832
polished_harmonic_regression_Fw	polished_model_development_registry	Fw	0.003230	0.003494	7.185
polished_periodic_mlp_Fw	polished_model_development_registry	Fw	0.089150	0.089207	195.822
polished_residual_harmonic_mlp_Fw	polished_model_development_registry	Fw	0.085646	0.085696	187.950
polished_tree_Fw	polished_model_development_registry	Fw	0.066315	0.066379	145.420
polished_periodic_mlp_harmonic_Fw	polished_model_development_registry	Fw	0.085830	0.085880	188.420
polished_temporal_convolution_Fw	polished_model_development_registry	Fw	0.088705	0.088763	195.075
polished_gru_sequence_Fw	polished_model_development_registry	Fw	0.087172	0.087229	191.682
polished_lstm_sequence_Fw	polished_model_development_registry	Fw	0.088000	0.088052	193.497
polished_periodic_temporal_convolution_Fw	polished_model_development_registry	Fw	0.085364	0.085443	187.236
polished_periodic_gru_sequence_Fw	polished_model_development_registry	Fw	0.086270	0.086329	189.713
polished_periodic_lstm_sequence_Fw	polished_model_development_registry	Fw	0.087733	0.087808	192.894
polished_residual_harmonic_gru_sequence_sparse_rcim_Fw	polished_model_development_registry	Fw	0.088164	0.088204	193.895
polished_residual_harmonic_gru_sequence_dense240_Fw	polished_model_development_registry	Fw	0.087700	0.087819	192.937
polished_residual_harmonic_gru_sequence_dense360_Fw	polished_model_development_registry	Fw	0.088752	0.089230	195.144
polished_residual_harmonic_lstm_sequence_sparse_rcim_Fw	polished_model_development_registry	Fw	0.087932	0.087972	193.422
polished_residual_harmonic_lstm_sequence_dense240_Fw	polished_model_development_registry	Fw	0.087313	0.087440	192.137
polished_residual_harmonic_lstm_sequence_dense360_Fw	polished_model_development_registry	Fw	0.088280	0.088701	194.256
polished_wave3_1_sequential_residual_offset_probe_Fw	polished_model_development_registry	Fw	0.088949	0.088999	195.544
polished_wave3_2_clean_sequential_residual_offset_Fw	polished_model_development_registry	Fw	0.087791	0.087843	193.000
polished_wave3_2_harmonic_residual_offset_Fw	polished_model_development_registry	Fw	0.087584	0.087638	192.572
polished_wave3_3_curve_aware_pointwise_control_Fw	polished_model_development_registry	Fw	0.087578	0.087631	192.719
polished_wave3_3_raw_centered_shape_curve_aware_Fw	polished_model_development_registry	Fw	0.084910	0.084970	186.769
polished_wave3_3_raw_offset_curve_aware_Fw	polished_model_development_registry	Fw	0.083671	0.083727	184.231
polished_wave3_3_full_curve_composite_Fw	polished_model_development_registry	Fw	0.085191	0.085256	187.347
polished_wave4_1_mae_robust_loss_Fw	polished_model_development_registry	Fw	0.084308	0.084363	185.612
polished_wave4_1_smooth_l1_robust_loss_Fw	polished_model_development_registry	Fw	0.085061	0.085122	187.116
polished_wave4_1_log_cosh_robust_loss_Fw	polished_model_development_registry	Fw	0.084634	0.084695	186.340
polished_wave4_2_quantile_p10_p50_p90_Fw	polished_model_development_registry	Fw	0.083077	0.083129	182.617
polished_wave4_2_gaussian_nll_Fw	polished_model_development_registry	Fw	0.083971	0.084023	184.509
polished_wave4_3_mixture_density_k2_Fw	polished_model_development_registry	Fw	0.085189	0.085240	187.358
polished_wave4_3_mixture_density_k3_Fw	polished_model_development_registry	Fw	0.083190	0.083242	182.913

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
polished_wave4_4_gru_latent_offset_residual_fw	polished_model_development_registry	Fw	0.088796	0.088857	195.229
polished_wave4_4_causal_tcn_latent_offset_residual_fw	polished_model_development_registry	Fw	0.087696	0.087760	192.689
polished_wave5_1_harmonic_prior_pointwise_control_fw	polished_model_development_registry	Fw	0.082881	0.082943	181.809
polished_wave5_1_harmonic_prior_smooth_l1_structured_fw	polished_model_development_registry	Fw	0.086190	0.086245	188.491

Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
harmonic_regression_fw	wave1_current_registry	Fw	0.003230	0.003494	7.185
periodic_mlp_harmonic_fw	wave1_periodic_mlp_harmonic_campaign	Fw	0.003254	0.003553	7.232
tree_fw	wave1_current_registry	Fw	0.066315	0.066379	145.420

Forward Reference Tree And Wave 2.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.066315	0.066379	145.420
temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.088705	0.088763	195.075
gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.087172	0.087229	191.682
lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.088000	0.088052	193.497
periodic_temporal_convolution_fw	wave2_temporal_entry_registry	Fw	0.085364	0.085443	187.236
periodic_gru_sequence_fw	wave2_temporal_entry_registry	Fw	0.086270	0.086329	189.713
periodic_lstm_sequence_fw	wave2_temporal_entry_registry	Fw	0.087733	0.087808	192.894

Forward Reference Tree And Wave 2.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.066315	0.066379	145.420
residual_harmonic_gru_sequence_sparse_rcim_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003194	0.003499	7.083
residual_harmonic_gru_sequence_dense240_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.006983	0.009275	15.722
residual_harmonic_gru_sequence_dense360_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007869	0.010574	17.740
residual_harmonic_lstm_sequence_sparse_rcim_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.003229	0.003533	7.164
residual_harmonic_lstm_sequence_dense240_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007042	0.009370	15.868
residual_harmonic_lstm_sequence_dense360_fw	wave2c_residual_harmonic_temporal_registry	Fw	0.007731	0.010235	17.430

Forward Reference Tree And Wave 3.1 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.066315	0.066379	145.420
wave3_1_sequential_residual_offset_probe_fw	wave3_1_offset_aware_probe_registry	Fw	0.003377	0.003799	7.487

Forward Reference Tree And Wave 3.3 Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.066315	0.066379	145.420
wave3_3_curve_aware_pointwise_control_fw	wave3_3_curve_aware_training_registry	Fw	0.003362	0.003612	7.474
wave3_3_raw_centered_shape_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.003174	0.003429	7.047
wave3_3_raw_offset_curve_aware_fw	wave3_3_curve_aware_training_registry	Fw	0.003269	0.003588	7.268
wave3_3_full_curve_composite_fw	wave3_3_curve_aware_training_registry	Fw	0.003251	0.003515	7.209

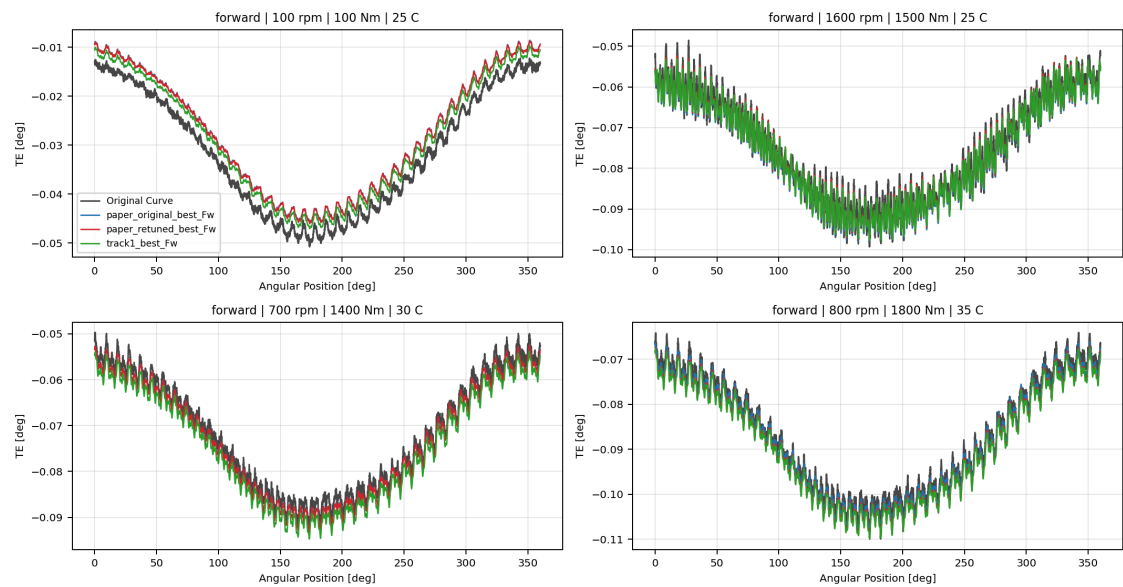
Forward Reference Tree And Wave 4 series Overlay

Candidate	Source	Surface	Curve MAE [deg]	Curve RMSE [deg]	Mean Error
paper_retuned_best_Fw	rcim_retuned	Fw	0.001839	0.002041	4.109
rcim_model_bank_reproduction_best_fw	rcim_model_bank_reproduction	Fw	0.003014	0.003204	6.819
tree_fw	wave1_current_registry	Fw	0.066315	0.066379	145.420
wave4_1_mae_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003134	0.003382	6.956
wave4_1_smooth_l1_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003300	0.003545	7.342
wave4_1_log_cosh_robust_loss_fw	wave4_1_robust_loss_registry	Fw	0.003344	0.003595	7.427

Comparison Gallery - Forward Reference Model Overlay

Included models: `paper_original_best_Fw` , `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` .

Forward Reference Model Overlay

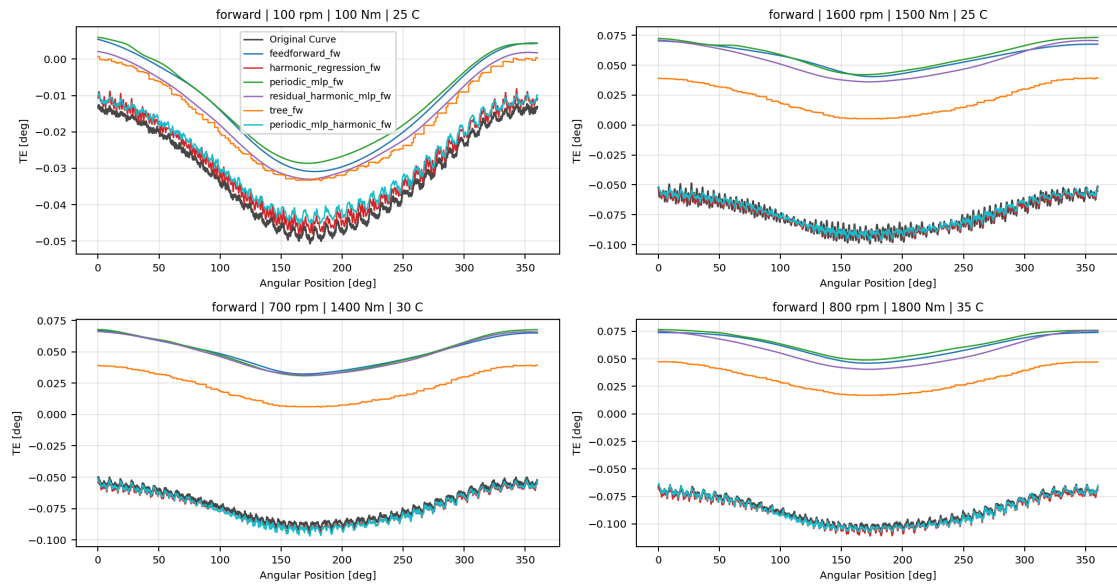


Forward Reference Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 1 Family Model Overlay

Included models: `feedforward_fw`, `harmonic_regression_fw`, `periodic_mlp_fw`, `residual_harmonic_mlp_fw`, `tree_fw`, `periodic_mlp_harmonic_fw`.

Forward Wave 1 Family Model Overlay

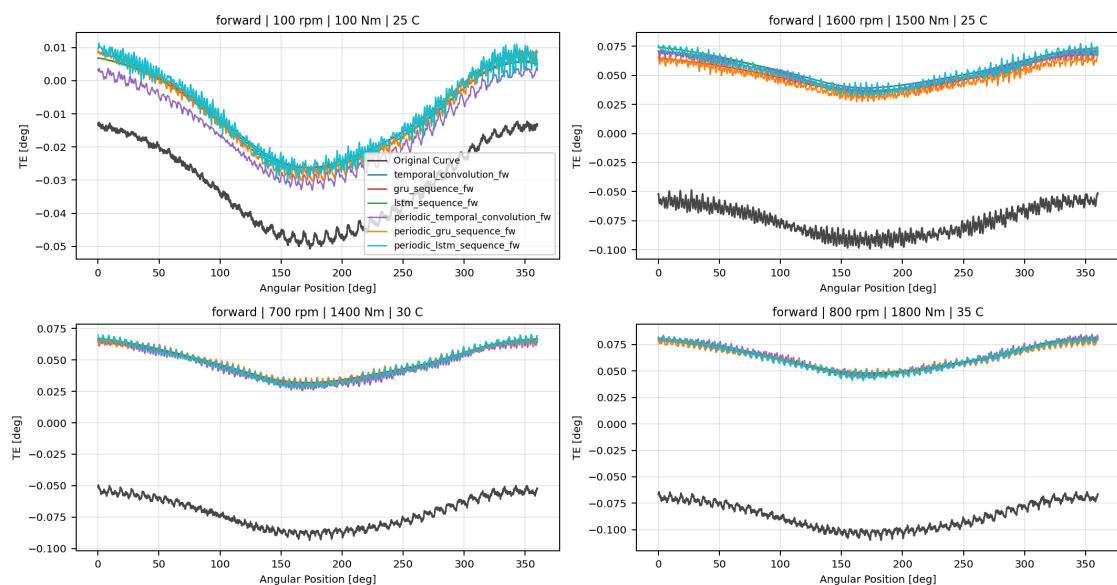


Forward Wave 1 Family Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 2.1 Temporal Model Overlay

Included models: `temporal_convolution_fw`, `gru_sequence_fw`, `lstm_sequence_fw`, `periodic_temporal_convolution_fw`, `periodic_gru_sequence_fw`, `periodic_lstm_sequence_fw`.

Forward Wave 2.1 Temporal Model Overlay

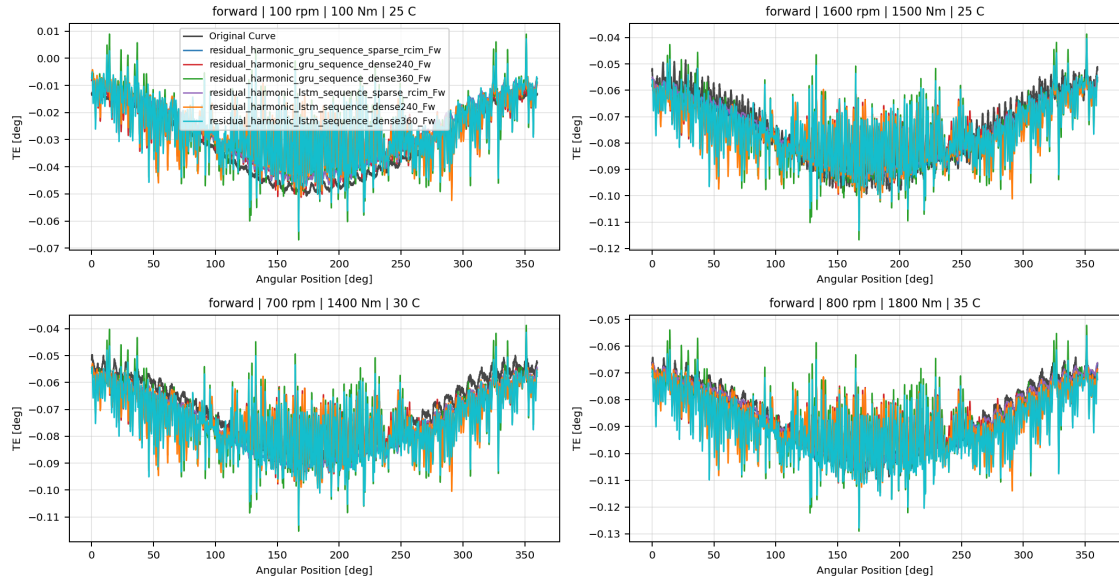


Forward Wave 2.1 Temporal Model Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 2.3 Residual Harmonic Temporal Overlay

Included models: `residual_harmonic_gru_sequence_sparse_rcim_Fw` ,
`residual_harmonic_gru_sequence_dense240_Fw` , `residual_harmonic_gru_sequence_dense360_Fw` ,
`residual_harmonic_lstm_sequence_sparse_rcim_Fw` , `residual_harmonic_lstm_sequence_dense240_Fw` ,
`residual_harmonic_lstm_sequence_dense360_Fw` .

Forward Wave 2.3 Residual Harmonic Temporal Overlay

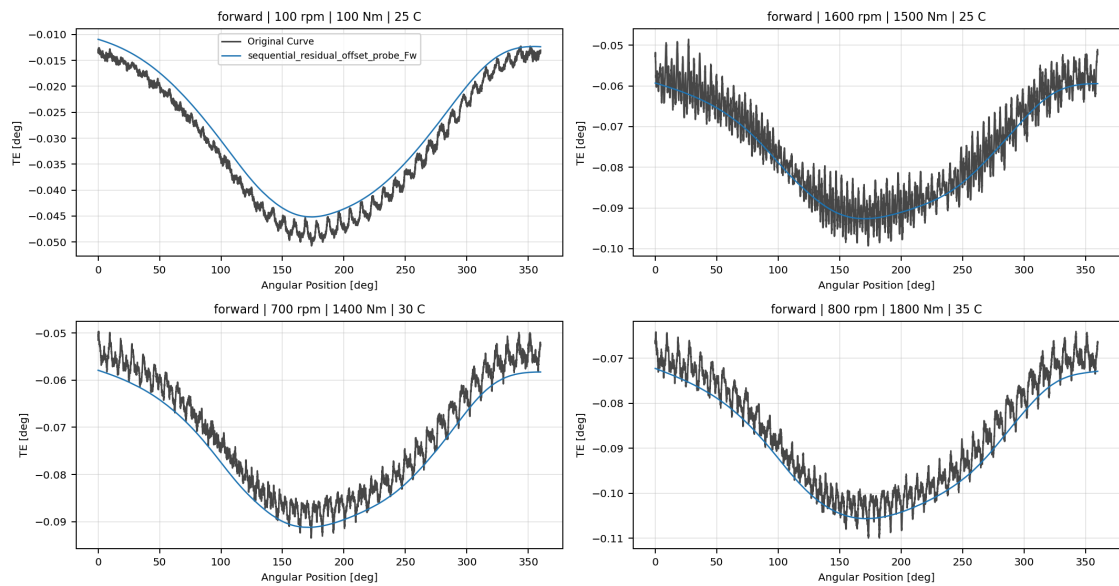


Forward Wave 2.3 Residual Harmonic Temporal Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 3.1 Offset-Aware Probe Overlay

Included models: `wave3_1_sequential_residual_offset_probe_fw` .

Forward Wave 3.1 Offset-Aware Probe Overlay

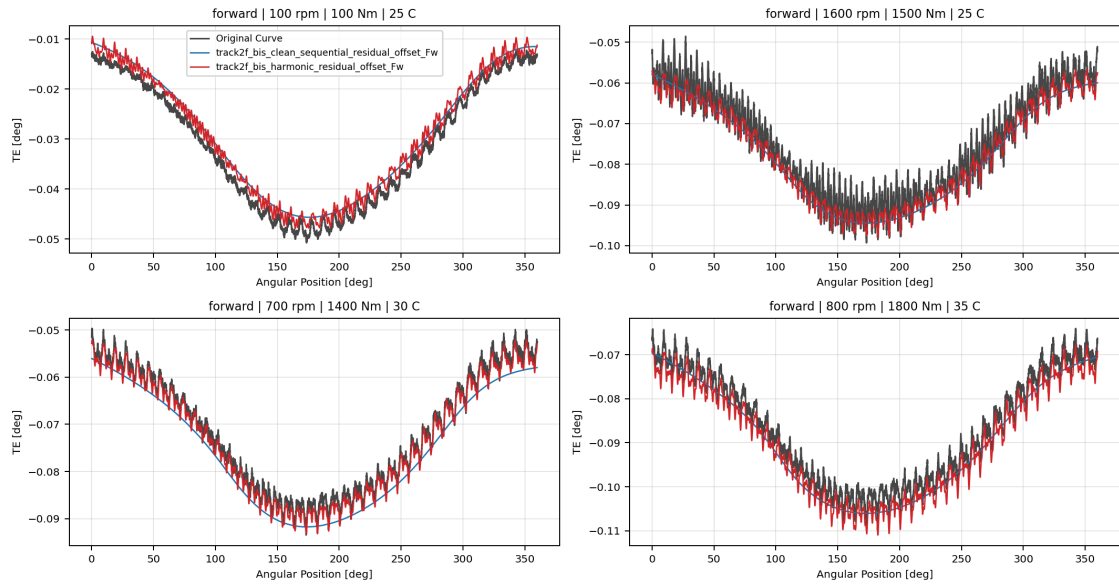


Forward Wave 3.1 Offset-Aware Probe Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 3.2 Harmonic-Offset Probe Overlay

Included models: `wave3_2_clean_sequential_residual_offset_fw` , `wave3_2_harmonic_residual_offset_fw` .

Forward Wave 3.2 Harmonic-Offset Probe Overlay

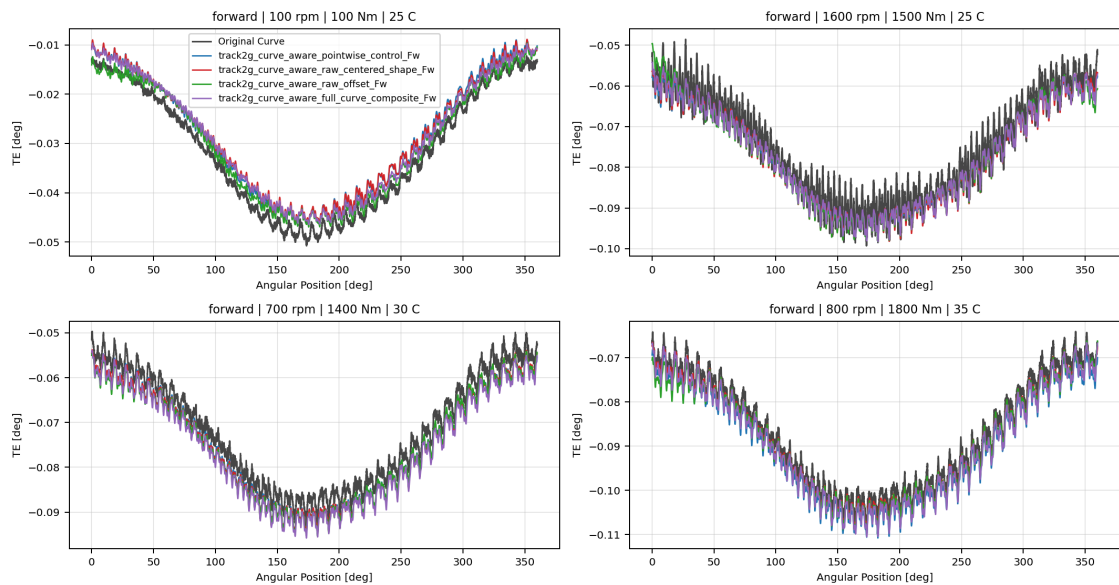


Forward Wave 3.2 Harmonic-Offset Probe Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 3.3 Curve-Aware Training Overlay

Included models: `wave3_3_curve_aware_pointwise_control_fw` , `wave3_3_raw_centered_shape_curve_aware_fw` , `wave3_3_raw_offset_curve_aware_fw` , `wave3_3_full_curve_composite_fw` .

Forward Wave 3.3 Curve-Aware Training Overlay

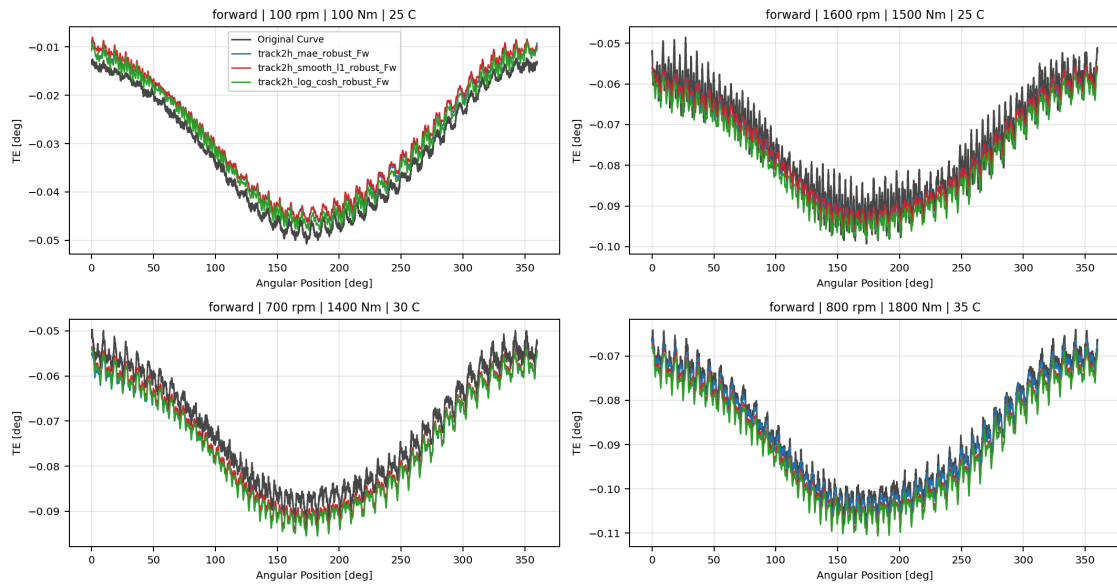


Forward Wave 3.3 Curve-Aware Training Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 4.1 Robust-Loss Overlay

Included models: `wave4_1_mae_robust_loss_fw` , `wave4_1_smooth_l1_robust_loss_fw` ,
`wave4_1_log_cosh_robust_loss_fw` .

Forward Wave 4.1 Robust-Loss Overlay

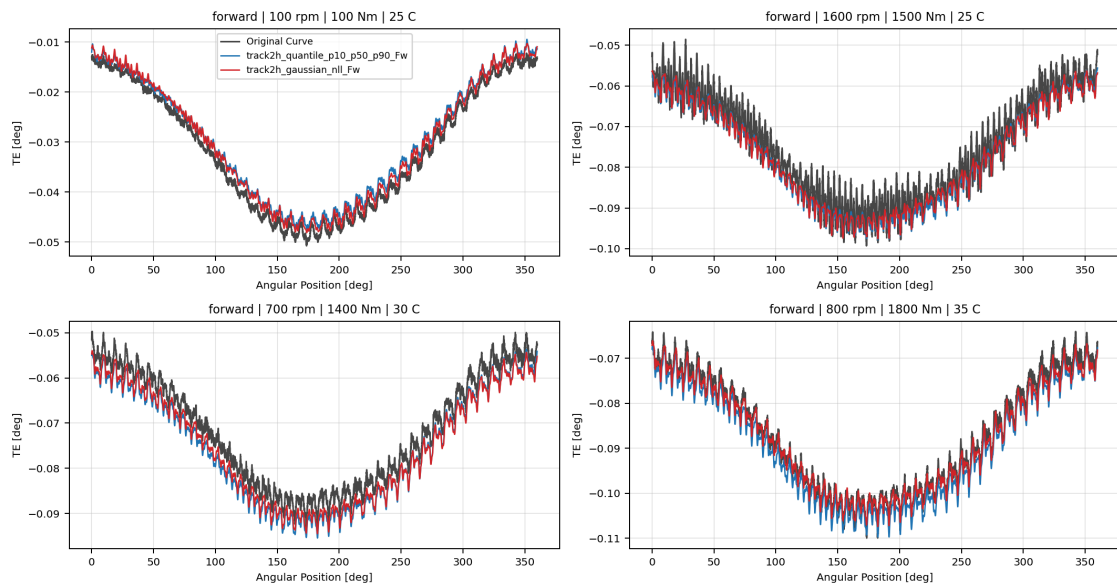


Forward Wave 4.1 Robust-Loss Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 4.2 Quantile Probabilistic Overlay

Included models: `wave4_2_quantile_p10_p50_p90_fw` , `wave4_2_gaussian_nll_fw` .

Forward Wave 4.2 Quantile Probabilistic Overlay

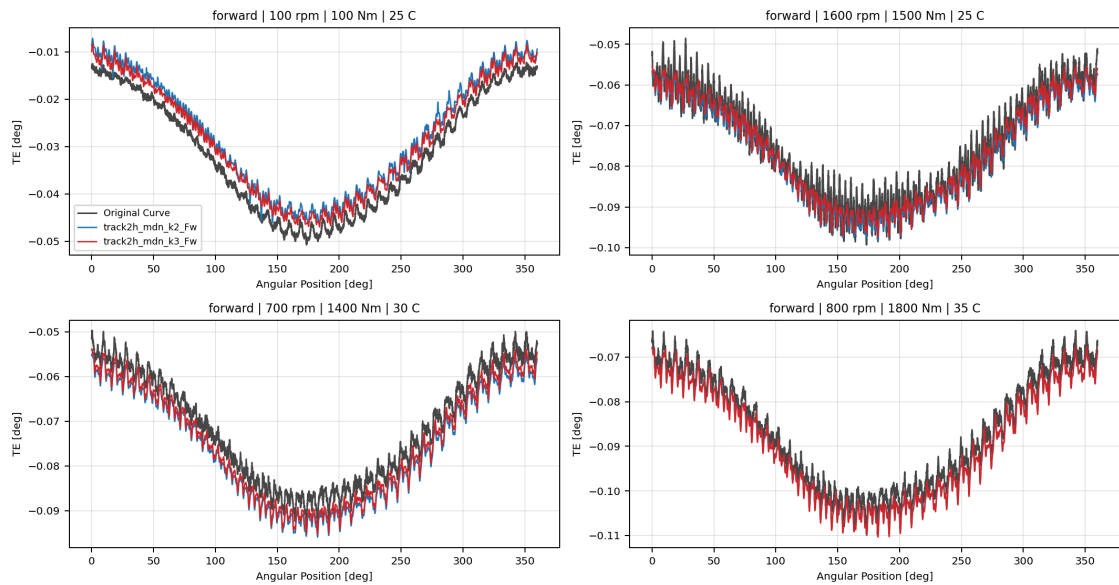


Forward Wave 4.2 Quantile Probabilistic Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 4.3 Mixture Density Overlay

Included models: `wave4_3_mixture_density_k2_fw` , `wave4_3_mixture_density_k3_fw` .

Forward Track2h Mixture Density Heads Registry Overlay

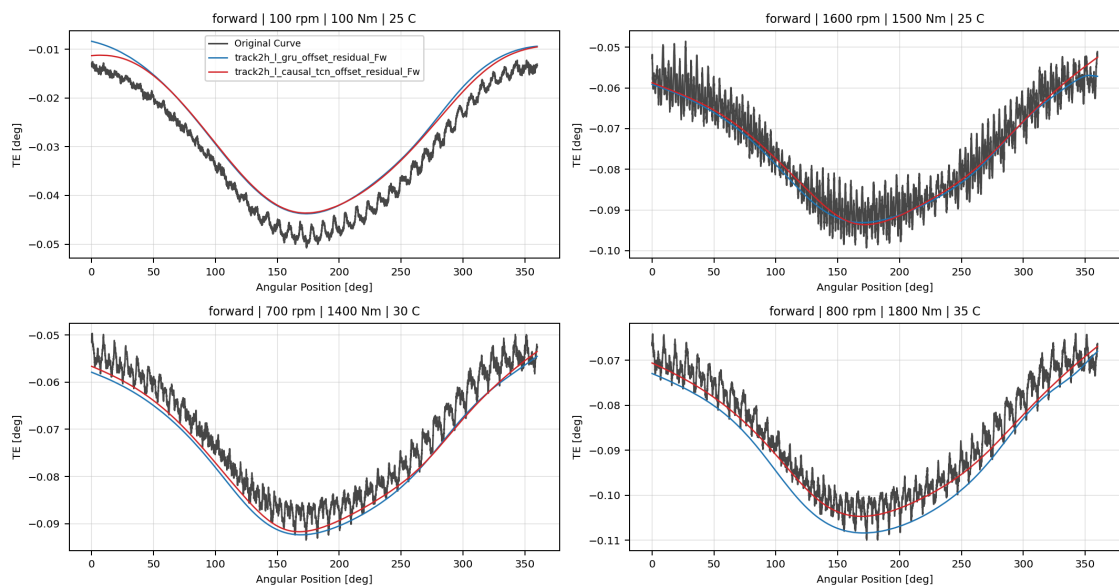


Forward Wave 4.3 Mixture Density Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 4.4 Latent State Hysteresis Overlay

Included models: `wave4_4_gru_latent_offset_residual_fw` , `wave4_4_causal_tcn_latent_offset_residual_fw` .

Forward Track2h Latent State Hysteresis Registry Overlay

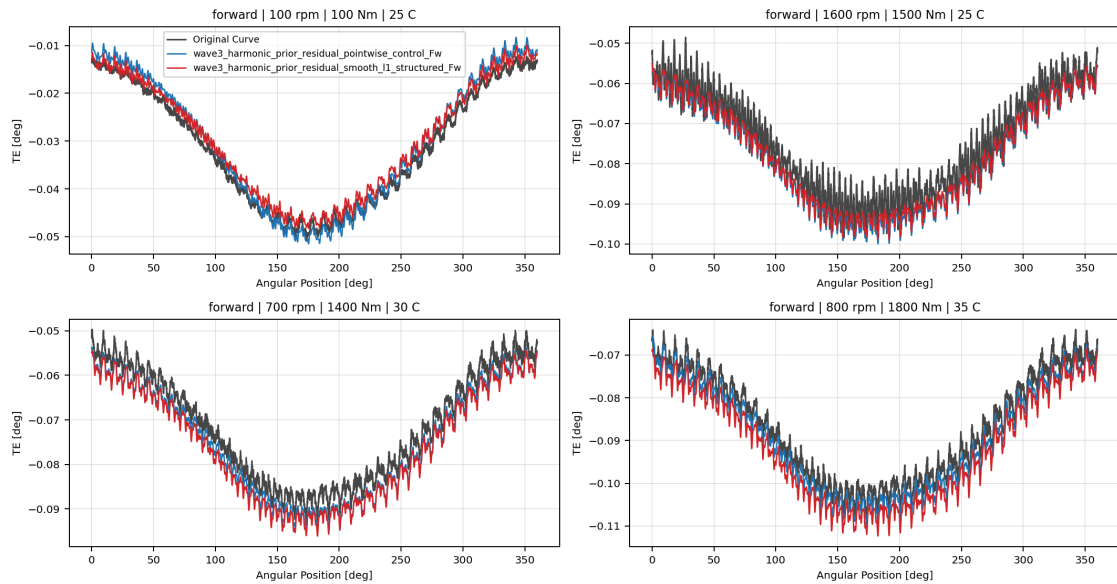


Forward Wave 4.4 Latent State Hysteresis Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave 3 Harmonic Prior Residual Registry Overlay

Included models: `wave5_1_harmonic_prior_pointwise_control_fw` ,
`wave5_1_harmonic_prior_smooth_l1_structured_fw` .

Forward Wave 3 Harmonic Prior Residual Registry Overlay

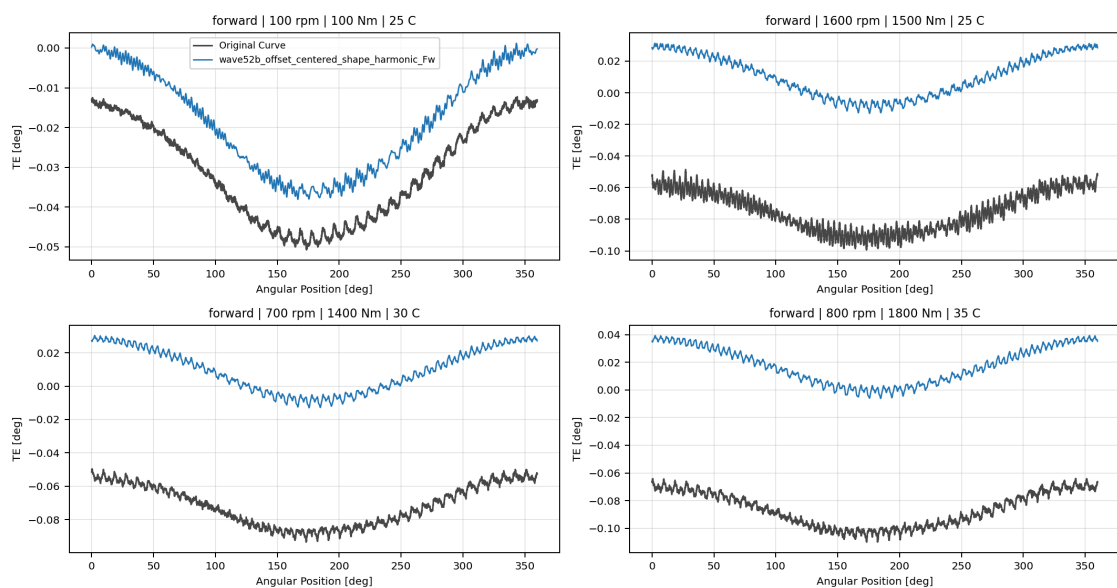


Forward Wave 3 Harmonic Prior Residual Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Wave52b Offset Harmonic Guided Registry Overlay

Included models: `wave52b_offset_centered_shape_harmonic_Fw` .

Forward Wave52b Offset Harmonic Guided Registry Overlay

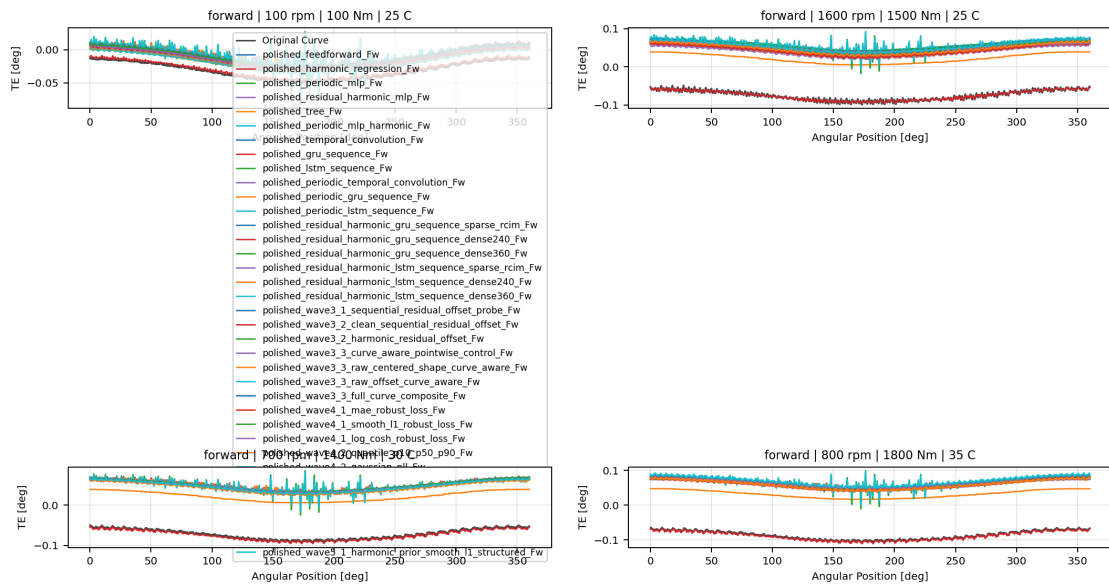


Forward Wave52b Offset Harmonic Guided Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Polished Model Development Registry Overlay

Included models: `polished_feedforward_Fw`, `polished_harmonic_regression_Fw`, `polished_periodic_mlp_Fw`, `polished_residual_harmonic_mlp_Fw`, `polished_tree_Fw`, `polished_periodic_mlp_harmonic_Fw`, `polished_temporal_convolution_Fw`, `polished_gru_sequence_Fw`, `polished_lstm_sequence_Fw`, `polished_periodic_temporal_convolution_Fw`, `polished_periodic_gru_sequence_Fw`, `polished_periodic_lstm_sequence_Fw`, `polished_residual_harmonic_gru_sequence_sparse_rcim_Fw`, `polished_residual_harmonic_gru_sequence_dense240_Fw`, `polished_residual_harmonic_gru_sequence_dense360_Fw`, `polished_residual_harmonic_lstm_sequence_sparse_rcim_Fw`, `polished_residual_harmonic_lstm_sequence_dense240_Fw`, `polished_residual_harmonic_lstm_sequence_dense360_Fw`, `polished_wave3_1_sequential_residual_offset_probe_Fw`, `polished_wave3_2_clean_sequential_residual_offset_Fw`, `polished_wave3_2_harmonic_residual_offset_Fw`, `polished_wave3_3_curve_aware_pointwise_control_Fw`, `polished_wave3_3_raw_centered_shape_curve_aware_Fw`, `polished_wave3_3_raw_offset_curve_aware_Fw`, `polished_wave3_3_full_curve_composite_Fw`, `polished_wave4_1_mae_robust_loss_Fw`, `polished_wave4_1_smooth_l1_robust_loss_Fw`, `polished_wave4_1_log_cosh_robust_loss_Fw`, `polished_wave4_2_quantile_p10_p50_p90_Fw`, `polished_wave4_2_gaussian_nll_Fw`, `polished_wave4_3_mixture_density_k2_Fw`, `polished_wave4_3_mixture_density_k3_Fw`, `polished_wave4_4_gru_latent_offset_residual_Fw`, `polished_wave4_4_causal_tcn_latent_offset_residual_Fw`, `polished_wave5_1_harmonic_prior_pointwise_control_Fw`, `polished_wave5_1_harmonic_prior_smooth_l1_structured_Fw`.

Forward Polished Model Development Registry Overlay

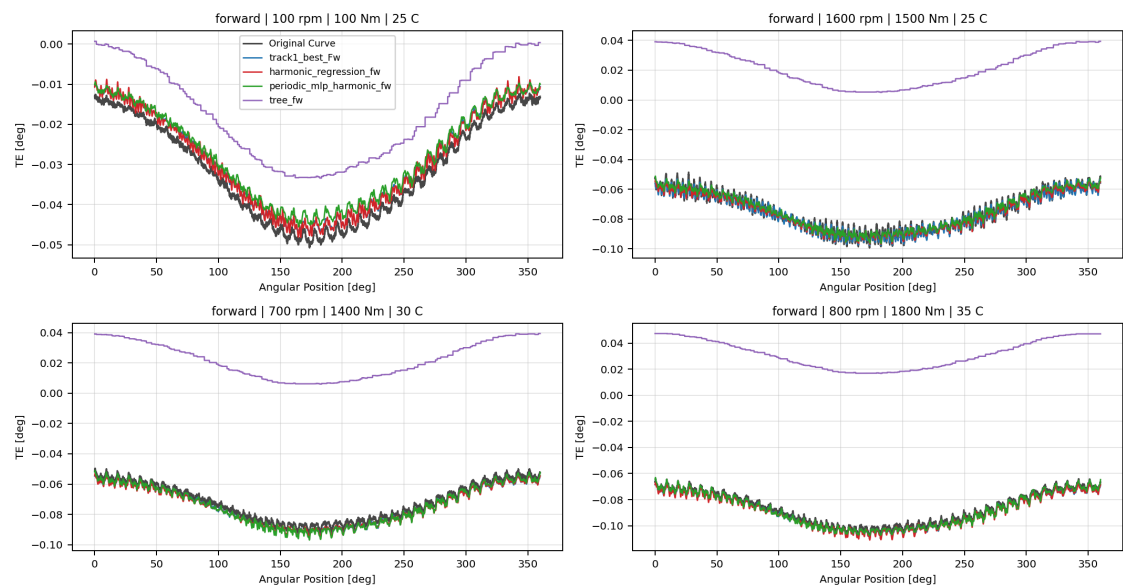


Forward Polished Model Development Registry Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

Included models: rcim_model_bank_reproduction_best_fw , harmonic_regression_fw , periodic_mlp_harmonic_fw , tree_fw .

Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay

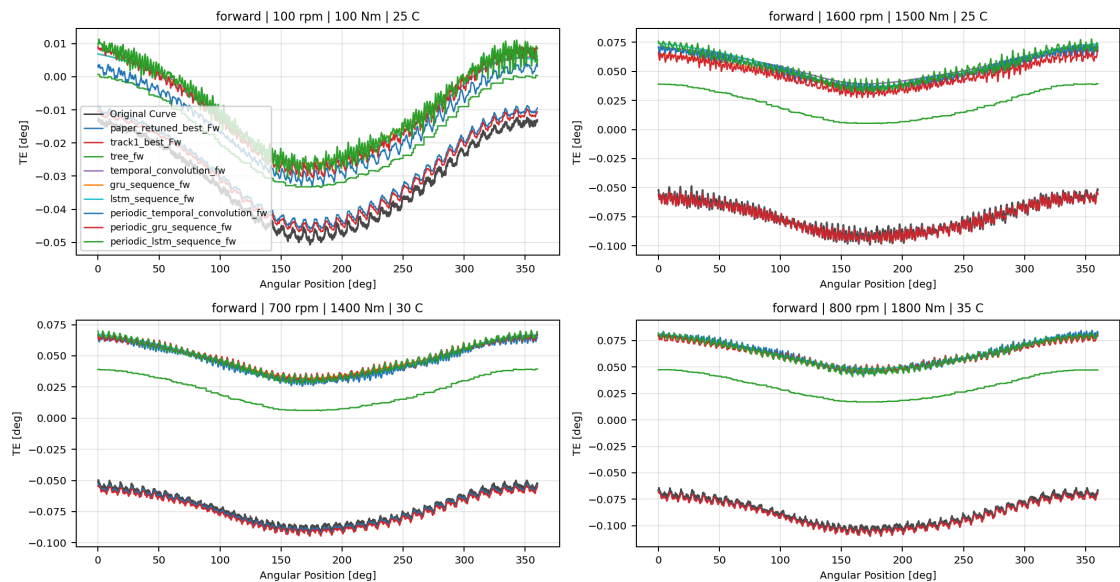


Forward RCIM Model-Bank Reproduction And Screened Wave 1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 2.1 Overlay

Included models: `paper_retuned_best_fw` , `rcim_model_bank_reproduction_best_fw` , `tree_fw` , `temporal_convolution_fw` , `gru_sequence_fw` , `lstm_sequence_fw` , `periodic_temporal_convolution_fw` , `periodic_gru_sequence_fw` , `periodic_lstm_sequence_fw` .

Forward Reference Tree And Wave 2.1 Overlay

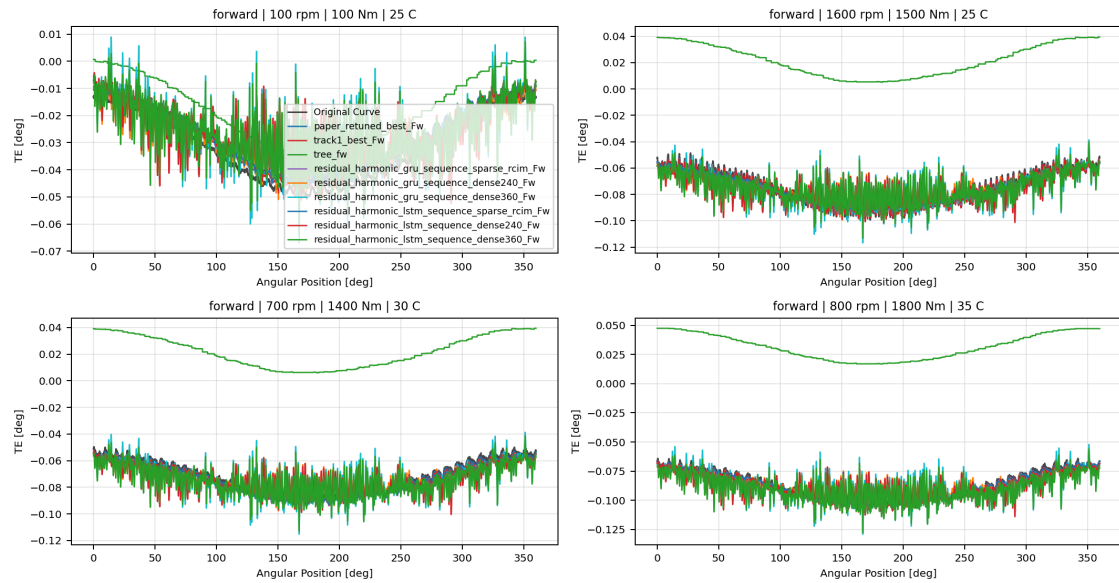


Forward Reference Tree And Wave 2.1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 2.3 Overlay

Included models: `paper_retuned_best_Fw`, `rcim_model_bank_reproduction_best_fw`, `tree_fw`, `residual_harmonic_gru_sequence_sparse_rcim_Fw`, `residual_harmonic_gru_sequence_dense240_Fw`, `residual_harmonic_gru_sequence_dense360_Fw`, `residual_harmonic_lstm_sequence_sparse_rcim_Fw`, `residual_harmonic_lstm_sequence_dense240_Fw`, `residual_harmonic_lstm_sequence_dense360_Fw`.

Forward Reference Tree And Wave 2.3 Overlay

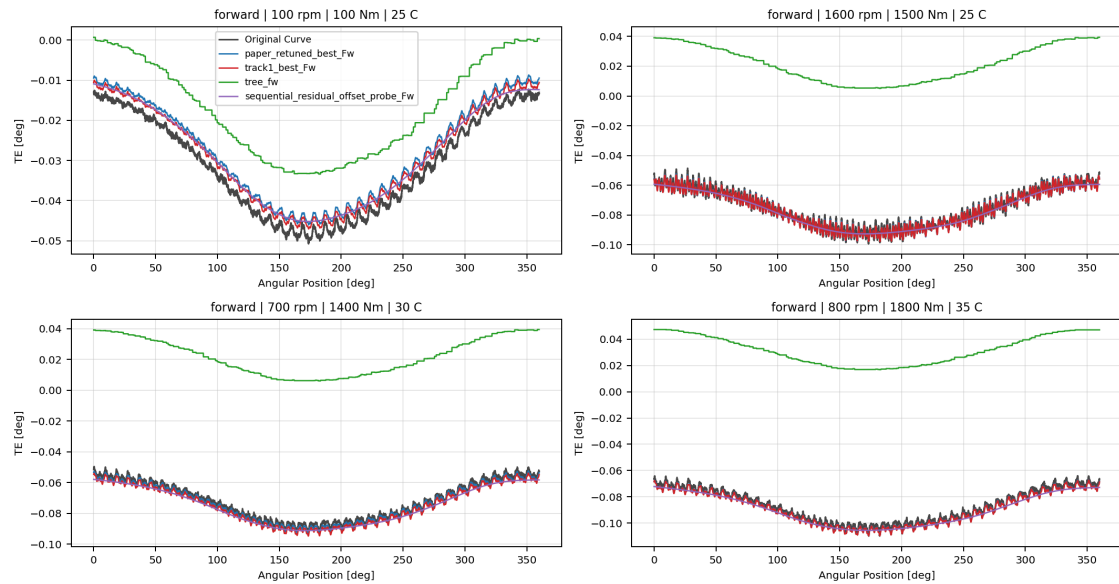


Forward Reference Tree And Wave 2.3 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 3.1 Overlay

Included models: `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` , `tree_fw` , `wave3_1_sequential_residual_offset_probe_fw` .

Forward Reference Tree And Wave 3.1 Overlay

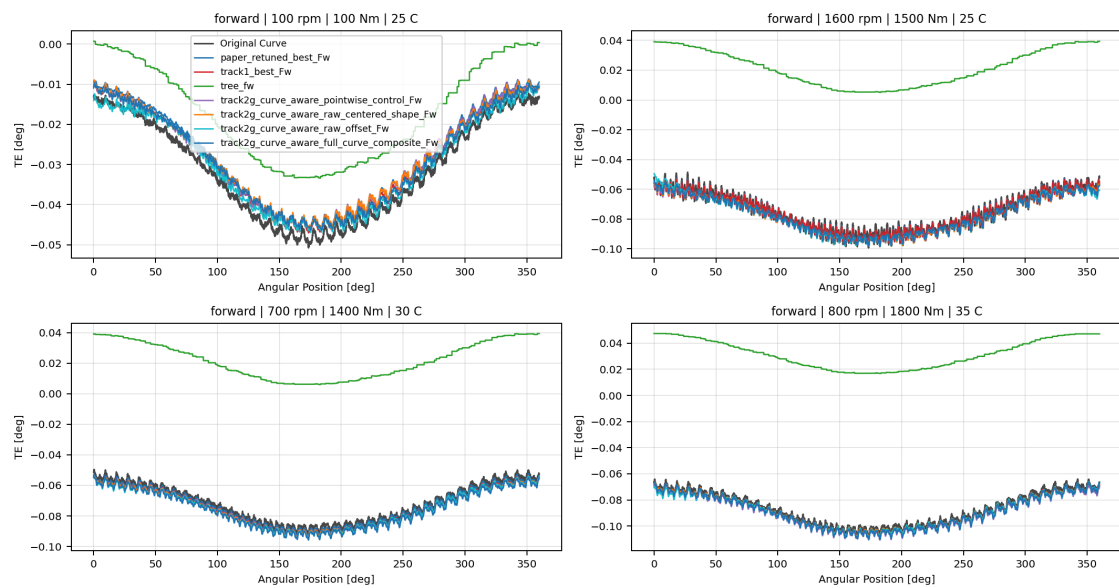


Forward Reference Tree And Wave 3.1 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 3.3 Overlay

Included models: `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` , `tree_fw` , `wave3_3_curve_aware_pointwise_control_fw` , `wave3_3_raw_centered_shape_curve_aware_fw` , `wave3_3_raw_offset_curve_aware_fw` , `wave3_3_full_curve_composite_fw` .

Forward Reference Tree And Wave 3.3 Overlay

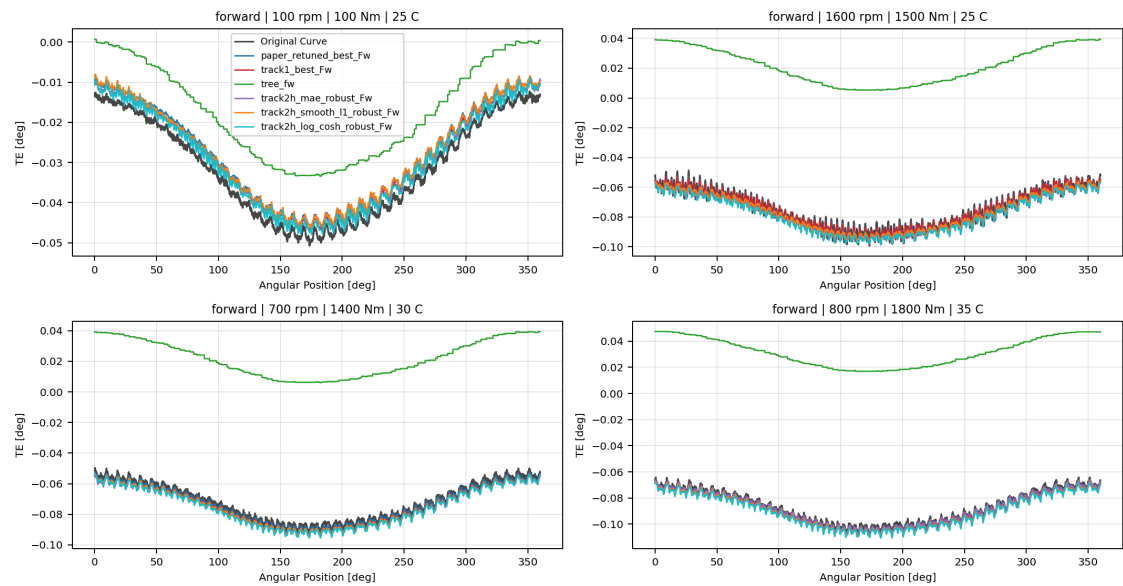


Forward Reference Tree And Wave 3.3 Overlay TE Curve Verification Pipeline comparison

Comparison Gallery - Forward Reference Tree And Wave 4 series Overlay

Included models: `paper_retuned_best_Fw` , `rcim_model_bank_reproduction_best_fw` , `tree_fw` , `wave4_1_mae_robust_loss_fw` , `wave4_1_smooth_l1_robust_loss_fw` , `wave4_1_log_cosh_robust_loss_fw` .

Forward Reference Tree And Wave 4 series Overlay



Forward Reference Tree And Wave 4 series Overlay TE Curve Verification Pipeline comparison

Output Artifacts

- output directory: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-05-11-32-46__track2_multi_model_curve_comparison_report` ;
- summary YAML: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-05-11-32-46__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_summary.yaml` ;
- metrics CSV: `output\validation_checks\track2_multi_model_curve_comparison_report\2026-07-05-11-32-46__track2_multi_model_curve_comparison_report\track2_multi_model_curve_comparison_metrics.csv` ;
- report Markdown:
`doc\reports\analysis\te_curve_verification_pipeline\02_visual_reports\dataset_surface_report\simplified_dataset\forward\overlay\[2026-07-04]\track2_multi_model_curve_comparison_report.md` .